1. **Healthcare & IoT Device Vulnerability Testbed**
2. **Team Members**

- Justin Bower - jbower2024@my.fit.edu
- Nathan Maloney - nmaloney2022@my.fit.edu
- Ipule Pipi - ipipi2022@my.fit.edu
- Jordan Chesley - jchesley2022@my.fit.edu

3. **Faculty advisor(s):** Dr. Sneha Sudhakaran & Dr. Abdullah Aydeger
4. **Client(s):** Security Researchers/Manufacturers/Engineers
5. **Progress of Current Milestone (Milestone 5 - October 26th)**

Task	% Complete	Justin	Nathan	Ipule	Jordan	To Do
DS18B20 I2C Sensor/Buspirate Research & Integration	50%	25%	0%	0%	25%	Execute VR/RE per Justin's research doc instructions
VR/RE & CVE database work	100%	60%	0%	0%	40%	Look for our own CVES now that known ones are documented
Containerized Deployments & Platform Integration/Security Testing	100%	50%	0%	0%	50%	Perform additional platform stability/security testing now that the div container/VM is deployable
Start-to-Finish RE & Local GUI Demo	100%	100%	0%	0%	0%	None
VM Mode Demo	100%	100%	0%	0%	0%	None
Project Management, Communication, & Documents	100%	100%	0%	0%	0%	Continue for next milestone

Cloud Platform Development, Collaborative Workspaces, UI/UX Redesign & Artifact Management	100%	0%	0%	100%	0%	Fully implemented
Network Traffic Analysis Script Updates & Integration (Wireshark/Tshark)	100%	50%	0%	0%	50%	Implement batch jobs/automation for NTA scripts
Firmware RE & Analysis Script Updates & Integration (Ghidra workflows, entropy analysis, Binwalk/Foremost segmentation)	100%	50%	0%	0%	50%	Implement batch jobs/automation for Firmware/RE analysis & exploitation scripts
Wi-Fi/Bluetooth Exploitation Script Updates & Integration	100%	50%	0%	0%	50%	Implement batch jobs/automation for Wi-Fi/Bluetooth Exploitation scripts
Local GUI Development	100%	50%	0%	0%	50%	None
Overall cloud improvements & Python notebook editor	100%	0%	100%	0%	0%	None
Cloud & Local Integration	50%	0%	40%	0%	0%	Implement firewall rules & deploy on a router

6. Discussion of each accomplished task (& obstacles) for Milestone 5:

- Task 1:

  Justin worked with Jordan to integrate the I2C sensor into the project using the Bus Pirate tool and PuTTY software. He determined the baud rate of 115200 & pinout, & with Jordan's help setting up, succeeded in getting connected on the /tty/USB0 serial interface. Next steps would be creating a small script that checks which serial connection is correct, or scripts for simple exploits, as outlined in Justin's research report.

- Task 2:

  Justin created a setup script for the VM version of the tool, which spins up a NixOS-based VM using QEMU. This provides the following benefits:

- Host Isolation: Sandboxes kernel-level exploits & malware inside the VM, keeping the physical host safe.
- Safe Network Recon: Grants NET_RAW/NET_ADMIN for packet sniffing without exposing your physical LAN.
- Cross-Platform Consistency: Guarantees an identical Linux kernel & Docker engine on macOS, Windows, or Linux.
- Zero-Trace Ephemeral State: Auto-destroys the VM & network bridges on exit, preventing host OS bloat.
- Secure GUI Pass-Through: Uses isolated X11 bridges so containers can't keylog or capture your host desktop.
- Instant Recovery: Nix-based immutable infrastructure allows users to "factory reset" the environment by just re-running the script.
- Frictionless Onboarding: Users only need the built-in NixOS & QEMU setup; moreover, all other dependencies/tools are baked into the VM/container stack.

- Task 3:

  Jordan utilized a dev container when building the local GUI for deployability and testing purposes. The dev container allows for an easily reproducible environment that can ship and build dependencies on run, as well as isolating network access.

- Task 4:

  Justin created the combined Local GUI and VM Mode demo and included it in the presentation.

- Task 5:

  Successfully implemented the collaborative workspace feature on the cloud platform, allowing multiple users to create & manage shared workspaces, collaborate on device

records, upload & access reverse-engineering artifacts, & execute analysis scripts within a shared environment.

Re-architected & redesigned major portions of the cloud platform's frontend, including the development of a new landing page, improved dashboard navigation, & redesigned workspace & device management interfaces to provide a more professional & intuitive user experience.

Addressed faculty advisor feedback by enhancing the artifact upload & management system to support different artifact categories & file types, allowing researchers to organize, locate, & access firmware, binary files, reports, logs, & other reverse-engineering outputs more efficiently.

Created device records & uploaded extracted reverse-engineering artifacts provided by the team, associating the files with their respective devices. Also populated the vulnerability database with identified CVE records & improved the interface for browsing & reviewing vulnerability information.

Major challenges included implementing workspace-based resource access while maintaining existing personal workspace functionality, ensuring that analysis jobs & uploaded artifacts were correctly associated with their respective workspaces & devices, & reorganizing the upload interface to support different artifact categories without disrupting existing functionality. These challenges were addressed through backend & frontend integration improvements, interface restructuring, & testing.

- Task 6:

  Up until this point in the project, Justin alone has completed approximately 60-90 percent of documents, including during semester 1. This is an obstacle; however, Jordan contributed much more to this milestone document. He also has assisted Justin with RE and local GUI development; prior to this point, Justin was the only one in the project with VR/RE experience. As a result, the documentation overhead has lessened slightly.

- Task 7:

  Nathan focused on cloud infrastructure and backend work: centralizing configuration into a shared file, updating Terraform and AWS permissions for multi-account deployment, testing deployment and existing services, adding Jupyter notebook support to the cloud script execution environment, and developing an API endpoint for incoming device traffic. Ipule focused on collaborative workspaces, UI/UX redesign, and artifact management: implementing shared workspaces with device/artifact management, member invitations, and access controls; redesigning the landing page, dashboard, and workspace interfaces; improving artifact upload/organization by category and file type;

creating device records and organizing reverse-engineering artifacts; and integrating/organizing CVE records with an improved vulnerability browsing interface. Collaborative workspaces, UI/UX redesign, and artifact management are explicitly attributed only to Ipule in the provided text.

- Task 8:

  Justin updated the network traffic analysis script from the first semester to fix bugs and ensure it integrates well with the launch mechanism of the Local GUI.

- Task 9

  Justin updated the firmware analysis/RE script from the first semester to fix bugs and ensure it integrates well with the launch mechanism of the Local GUI.

- Task 10

  Justin updated the Wi-Fi/bluetooth network analysis script from the first semester to fix bugs and ensure it integrates well with the launch mechanism of the Local GUI.

- Task 11

  Jordan got the Local GUI in-frame display for all tools working with some initially kludgy implementations. In addition, he made the dropdown menu for launching tools provide recommendations based upon selected file type & made a launch parameters field to allow for launch flags parameters to be used when choosing a tool.

  However, when transitioning the Local GUI back to NixOS on the lab computer, exposed/resulted in obstacles that needed to be overcome. First, Jordan & Justin struggled with versions/package differences in the Tkinter Python module. In addition, Justin needed to resolve an X11 display error that popped up when launching Ghidra in the GUI mode. He fixed this by reducing hardcoded file paths for necessary dependencies in NixOS through implementing dynamic pathing & better error handling in the Local GUI Python files. In addition, he added necessary permissions & settings in the NixOS config file.

- Task 12

  Nathan updated the cloud execution environment with necessary Jupyter packages and added functionality to create and execute notebooks, integrated with existing cloud scripting. He also tested notebook execution with Python code, verified outputs and no interference with existing scripting, and resolved configuration/integration issues found during testing.

- Task 10:

  Several changes were made to the cloud backend to improve its organization, flexibility, & overall management. One of the main goals of these changes was to make it easier to transfer & deploy the cloud infrastructure to a separate AWS account. This allows the project to operate in different AWS environments without requiring significant modifications to the existing code or infrastructure. These improvements also make the backend easier to maintain as additional features & services are introduced.

  A major part of this process involved centralizing various configuration settings & values into a single configuration file. Previously, some of these values were defined in different locations throughout the project, making changes more difficult & increasing the possibility of configuration errors. By moving these settings into one location, it is now easier to manage account-specific information, adjust deployment settings, & maintain consistency across the different components of the cloud infrastructure. This also reduces the amount of manual work required when deploying the project to a new AWS account.

  Another significant addition was support for creating & running Jupyter notebooks within the cloud environment. This feature provides users with a more flexible way to perform analysis by allowing them to write & execute Python code directly in the cloud. Jupyter notebooks are particularly useful for cybersecurity research because they allow users to combine code, analysis results, & documentation in a single environment. This creates opportunities to develop and test custom analysis scripts without requiring users to configure all necessary tools on their local machines.

  Unlike the existing cloud scripting capabilities, which already allow users to execute analysis scripts, Jupyter notebooks provide a more interactive approach to working with code & results. Rather than running an entire script at once, users can execute individual sections of code, inspect the output, & make adjustments without restarting the entire analysis process. This can be especially useful when experimenting with firmware analysis techniques or investigating unexpected results, since users can make changes & immediately observe their effects.

- Task 11:

  Research was conducted into the router's administrative interface to determine how network traffic could be restricted to a specific URL. The goal was to identify the settings & capabilities needed to block all other outgoing traffic while allowing devices connected to the testbed to communicate with a designated API endpoint. This involved investigating the router's firewall settings, access control features, & traffic filtering capabilities. The router has not yet been configured, as this stage focused on researching the available options & determining how the restrictions could be implemented. Additionally, an API was developed to receive traffic sent to the designated endpoint.

This provides a centralized destination for communications from devices connected to the testbed. Together, the router research & API development establish the groundwork for controlling network communications & ensuring that testbed devices only send traffic to approved destinations.

7. Discussion of contribution of each team member to Milestone 5:

Justin Bower:

- Maintained email communications with advisor & communicated their comments, suggestions, & requests to the rest of the team
- Created, shared, & did the most work for project documents
- Read advisor's research papers, & did other online research, then created a write-up of said research. Worked with Jordan to integrate the I2C sensor into the project using the bus-pirate tool. He determined the baud rate of 115200 & pinout, & with Jordan's help setting up, succeeded in getting connected on the /tty/USB0 serial interface.
- Extracted the remaining binary files for the Wi-Fi chipset from the open-source Linux driver using the reproducible workflow (script). Then, he uploaded them to GitHub for Ipule to associate with their respective devices. Since then, he has separated this process from the main NixOS config file into a standalone script.
- Worked with Jordan to verify Local GUI functionality in preparation for the start-to-finish demo. In his testing, he found that he needed to fix hardcoded paths to dynamic methods for linking with the GUI-based tools. This work was successful & made the tool more robust.
- Took the Network Traffic Analysis scripts from semester 1, adapted them to work with the Local GUI, & tested them.
- Took the Wi-Fi/Bluetooth Exploitation scripts from semester 1, adapted them to work with the Local GUI, & tested them.
- Took the Firmware Analysis/Exploitation scripts from semester 1, adapted them to work with the Local GUI, & tested them.
- Successfully created a start-to-finish RE/VR demo using Wi-Fi chipset binary files.
- Successfully created a demo of the VM version of the tool, which sets up a NixOS-based VM using QEMU. This provides the following benefits:
- Host Isolation: Sandboxes kernel-level exploits & malware inside the VM, keeping the physical host safe.
- Safe Network Recon: Grants NET_RAW/NET_ADMIN for packet sniffing without exposing your physical LAN.
- Cross-Platform Consistency: Guarantees an identical Linux kernel & Docker engine on macOS, Windows, or Linux.
- Zero-Trace Ephemeral State: Auto-destroys the VM & network bridges on exit, preventing host OS bloat.

- Secure GUI Pass-Through: Uses isolated X11 bridges so containers can't keylog or capture your host desktop.
- Instant Recovery: Nix-based immutable infrastructure allows users to "factory reset" the environment by just re-running the script.
- Frictionless Onboarding: Users only need the built-in NixOS & QEMU setup; moreover, all other dependencies/tools are baked into the VM/container stack.

Nathan Maloney:

- Reviewed existing AWS infrastructure to identify account-specific settings/values, centralized scattered backend and infrastructure configuration into a shared config file, and updated backend components to reference it instead of hardcoded values.
- Adjusted Terraform configurations for deployment across different AWS accounts and updated AWS permissions/deployment settings for the new environment.
- Tested cloud infrastructure deployment using the updated configurations and verified existing backend services continued functioning.
- Reviewed the existing cloud script execution system and identified the additional dependencies/configuration required to support Jupyter notebooks.
- Updated the cloud execution environment with necessary Jupyter packages and added functionality to create and execute notebooks, integrated with existing cloud scripting.
- Tested notebook execution with Python code, verified outputs and no interference with existing scripting, and resolved configuration/integration issues found during testing.
- Researched the router's administrative interface and configuration options, including firewall rules and access-control settings for restricting outgoing traffic.
- Explored methods to block all outgoing traffic except communications to a designated URL/endpoint, while preserving necessary testbed functionality.
- Identified potential limitations of the router's traffic-filtering capabilities.
- Developed an API endpoint and basic request-handling structure to receive incoming traffic from connected devices.
- Planned integration between the router's traffic restrictions and the API, and identified remaining configuration/testing steps required to implement the restrictions.

Ipule Pipi:

- Collaborative Workspace Implementation: Designed & implemented the collaborative workspace feature on the cloud platform, allowing multiple users to collaborate within shared workspaces. Users can create & manage devices, upload & access associated artifacts, & execute analysis scripts within a shared workspace environment. Implemented workspace management, member invitations, & workspace-based access controls.
- Cloud Platform UI/UX Redesign: Re-architected & redesigned major portions of the cloud platform's frontend to improve usability, navigation, & overall user experience.

Developed a new professional landing page, redesigned the dashboard & workspace interfaces, & improved the organization & consistency of the platform's UI components.

- Artifact Upload & File Organization Improvements: Addressed feedback from Dr. Sudhakaran by improving the artifact upload & management functionality. Implemented support for organizing & browsing uploaded artifacts by category & file type, making it easier for researchers to locate firmware, binary files, logs, reports, & other reverse-engineering outputs. Improved the upload workflow to allow artifacts to be uploaded directly from their associated device pages.
- Reverse-Engineering Artifact Integration: Created device records on the cloud platform for the devices investigated by Justin & Jordan. Uploaded & organized the extracted reverse-engineering files & associated them with their corresponding device records, making the artifacts accessible through the platform for further analysis.
- Vulnerability Database Integration: Entered & organized identified Common Vulnerabilities & Exposures (CVE) records within the cloud platform's vulnerability database. Improved the vulnerability browsing interface to make it easier for users to search, review, & access detailed vulnerability information.
- Reverse-Engineering Assistance: Assisted the reverse-engineering team with the physical disassembly & examination of a heart-monitoring device. Participated in the hardware investigation process to gain practical understanding of device internals & the firmware extraction workflow. Direct firmware extraction & network traffic capture were not completed as part of this contribution.

Jordan Chesley:

- Connected the I2C sensor to the Bus Pirate for serial interfacing.
- Made graphical applications launch inside the Local GUI by passing the physical device's display into the dev container using Xephyr and the X11 API.
- Added a PTY to the Local GUI for CLI tools so that ANSI escape sequences can be processed and for better interaction with the tools.
- Added file selection UI elements to the Local GUI so users can dynamically select files to launch with tools.
- Added an optional arguments text field to the Local GUI so users can pass any additional command-line arguments to their tools.
- Created a tool plugin API for the Local GUI so that tools/scripts can be programmatically added with ease. The API allows users to define elements of a tool, such as the tool name, the description, the launch command and the command's structure, the tool category, and the file types associated with the tool.
- Modified the Tools dropdown in the Local GUI so that tools are listed by category and recommended based on the selected file. When a file is selected in the UI, the dropdown filters tools by their associated file types and lists those associated tools at the beginning of the dropdown. Other tools are then sorted by their category.

- Gave the Local GUI a dark theme.
- Allowed for multiple tools to be run at the same time. Open tools can be navigated through a tab menu above the tool frame.
- Began work on entropy analysis/file carving scripts.

8. Plan for the next Milestone (Milestone 6 - Nov 23rd)

Task	Justin	Nathan	Ipule	Jordan
Finalize CVE & RE Documentation	Finalize CVE entries & RE documentation for the final demo.	Assist with finalizing CVE entries & RE documentation.	Assist with finalizing CVE entries & RE documentation.	Assist with finalizing CVE entries & RE documentation.
Cyber Tooling Demos	Add demos of the cyber tooling into the final presentation.	None	None	Assist with preparing demos of the cyber tooling integrations.
Project Management & Comms	Keep team moving; maintain constant email/Discord communications with team & Professors.	None	None	None
AWS Backend Security & Integration	None	Finalize AWS permissions & backend security (least-privilege). Complete integration between local tools and cloud services	Assist Nathan	None
Dev Container Security Testing	Assist with testing.	Assist with testing.	Assist with testing.	Finalize & test containerized cloud analysis & isolation. Perform testing for final demo.

Local RE Pipelines Security Testing	Assist with testing.	Assist with testing.	Assist with testing.	Conduct testing of tool function security
End-to-End Web Workflow	None	Assist Ipule	Complete web workflow: device registration -> firmware upload -> analysis execution -> result retrieval.	None
Dashboard Integration & UX	None	Assist Ipule	Integrate RE/vulnerability outputs (entropy, discovered files, CVEs) into the dashboard. Organize devices, firmware, jobs, & outputs into a clear workflow.	None
Web App Edge-Case Testing	None	Assist Ipule	Perform edge-case testing (auth, ownership, upload retry, analysis initiation, results). Resolve integration issues & improve error handling.	None
Finalize Local GUI & Exploitation Scripts Batching Automation	None	Implement exploitation scripts batching & automation	Local RE pipelines by feeding results to the web backend.	None
Poster: Overall Design & Layout	Lead layout design & formatting; consolidate all sections into the	Assist Justin with layout design, formatting, & consolidating sections.	Assist Justin with layout design, formatting, & consolidating sections.	Assist Justin with layout design, formatting, &

	final poster template.			consolidating sections.
Poster: Content Drafting & Visuals	Draft the RE/VR methodology, local GUI, & cyber tooling sections; provide workflow diagrams.	Draft cloud architecture & backend integration sections; provide diagrams/screenshots.	Draft web UI/UX & collaborative workspace sections; provide screenshots of the dashboard.	Draft the local pipeline & exploitation/analysis sections; provide screenshots of the Local GUI.
Poster: Overall Review & Polish	Proofread for consistency & formatting; ensure all required sections are present.	Verify technical accuracy of cloud/backend claims & diagrams.	Verify technical accuracy of web/dashboard claims & visuals.	Assist Justin with proofreading, formatting, & final polish.

9. Discussion of each planned task for Milestone 6:

- Task 1: Finalize CVE & RE Documentation
  We've completed disassembling devices and researching publicly disclosed vulnerabilities for the chipsets of our devices. We will perform our own research on the chipsets to discover potentially undisclosed vulnerabilities and log them on our platform.
- Task 2: Cyber Tooling Demos
  Create demos of the Local GUI and Cloud platform. Once the two separate platforms are connected with a middleman API, showcase the functionality of the Local GUI sending data to the Cloud platform.
- Task 3: AWS Backend Security & Integration
  Review the security of the AWS infrastructure, making sure accounts and infrastructure have the least privileges possible to prevent data leaks, unauthorized requests, etc. Additionally, complete the integration of local tools with cloud services.
- Task 4: Dev Container Security Testing
  Test containerized cloud analysis and ensure that cloud containers are properly isolated and secure from unauthorized access.
- Task 5: Local RE Pipelines Security Testing
  Test the RE pipelines and ensure files and networks outside of the container cannot be accessed.
- Task 6: End-to-End Web Workflow
  Complete and test the full project workflow on the cloud platform. Users register a

device, upload the device firmware, have the firmware analyzed on the cloud platform, and retrieve the resulting analysis files.

- Task 7: Dashboard Integration & UX
  Integrate uploaded data (firmware, carved files, logs, network scans, etc.) into the website dashboard.
- Task 8: Web App Edge-Case Testing
  Test the cloud platform for potential unhandled edge cases and implement fixes/handles for them.
- Task 9: Finalize Local GUI & Exploitation Scripts Batching Automation
  Implement the Launch Options dropdown to launch either local, remote, or simulated, as well as create and add the automated scripts into the Tools dropdown.
- Task 10: Poster: Overall Design & Layout
  Design the layout of the project poster. Identify the major sections that should be included in the poster and accompanying visuals and determine a layout to best fit the components.
- Task 11: Poster: Content Drafting & Visuals
  Upon completion of Task 10, begin drafting the poster and its contents. Each team member will focus on a particular section of the poster to draft and will also create an accompanying visual for their section.
- Task 12: Poster: Overall Review & Polish
  Upon completion of Task 11, work with the client (Dr. Sudhakaran) to review and make recommended revisions to the poster.

10. Date(s) of meeting(s) with Client during the current milestone:

9/30/26 @ 5:00 p.m., 10/23/26 @ 11:00 a.m., 10/26/27 @ 11:00 a.m.

11. Client feedback on the current milestone

• See Faculty Advisor Feedback below

12. Date(s) of meeting(s) with Faculty Advisor during the current milestone:

9/30/26 @ 5:00 p.m., 10/23/26 @ 11:00 a.m., 10/26/27 @ 11:00 a.m.

13. Faculty Advisor feedback on each task for the current Milestone

• Task 1:

• Task 2:

• Task 3:

• Task 4:

• Task 5:

Faculty Advisor Signature: _____ Date: _____